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**B.Sc. (E.C.S.) (Semester - III) (New) (NEP CBCS) Examination:
October/November – 2025
Database Management Systems (G06-DSC1-0302)**

Max. Marks: 30

- Instructions:** 1) All questions are compulsory.
2) Draw neat diagrams and give equations wherever necessary.
3) Figures to the right indicate full marks.

06

- 1) In a 3-tier DBMS architecture, the middle layer is also called: _____.
 - a) Internal level
 - b) External level
 - c) Conceptual level
 - d) Physical level
- 2) Which normal form is concerned with multi-valued dependency?
 - a) 3NF
 - b) 4NF
 - c) 5NF
 - d) BCNF
- 3) Which command is used to remove all rows from a table but keep the structure?
 - a) DROP
 - b) DELETE
 - c) TRUNCATE
 - d) RENAME
- 4) Which type of JOIN returns all rows from both tables, combining matching rows?
 - a) Inner Join
 - b) Left Join
 - c) Right Join
 - d) Full Outer Join
- 5) What is the purpose of the FOREIGN KEY in a table?
 - a) To uniquely identify a row
 - b) To enforce referential integrity
 - c) To speed up queries
 - d) To store images
- 6) What happens when a base table used in a view is dropped?
 - a) The view is automatically converted into a table
 - b) The view becomes invalid
 - c) The view remains active with old data
 - d) Nothing happens

Q.2 Answer the following. (Any Three) **06**

- a) Define Instance and scheme of database.
- b) List out built in string function.
- c) Define generalization and specialization in ER modeling.
- d) State any two disadvantages of file based system.
- e) Define Data -Independence.

Q.3 Answer the following. (Any Two) **06**

- a) Differentiate DBMS and RDBMS.
- b) Discuss the purpose of rollback, commit, and savepoint in DCL.
- c) Explain group by clause with an example.

Q.4 Answer the following. (Any Two) **06**

- a) Explain foreign key constraints with example.
- b) Explain special operator:
 - 1) IN.
 - 2) Between.
 - 3) Like.
- c) What is View? Explain View for DML operation.

Q.5 Answer the following. (Any One) **06**

- a) Explain 1NF, 2NF, and 3NF with suitable examples.
- b) Consider the following **Sales table** and write queries:

OrderID	Customer	Category	Sales	Profit	Region	Order Date
101	Jay	Furniture	1500	300	North	2020-05-10
102	Mahesh	Stationery	500	120	South	2021-01-15
103	Ramesh	Electronics	2500	400	East	2020-08-20
104	Priya	Furniture	1200	200	West	2021-06-11
105	Ankita	Stationery	800	150	North	2021-09-05
106	Dinesh	Furniture	1700	250	East	2020-11-23

1. Find total sales by region.
2. Add new column which extracts the **Month** from Order Date.
3. Count all orders which is in year 2021.

Perform following queries on Emp and Sales table respectively.

Emp ID	Name	DeptID	DeptID	DeptName
1	JayKumar	10	10	HR
2	Mangesh	20	20	Marketing
3	Aarati	Null	30	IT

4. Which join operation to perform that returns matching dept name.
5. Returns all employees, even if no matching department.
6. Returns all departments, even if no employee in it.